

EDUCATION

University of California, Los Angeles (UCLA)	M.S. Aerospace Engineering	09/2026 – 06/2028
San Diego State University (SDSU)	B.S. Aerospace Engineering (Physics Minor)	08/2023 – 05/2026
Santa Monica College (SMC)	A.A. General Science (Honors)	06/2020 – 07/2023

ENGINEERING EXPERIENCE

Undergraduate Research Assistant (GNC)	SDSU SPACE Lab	11/2023 – 08/2024
- Developed Python autonomous drone navigation algorithms using OpenCV, enabling QR-code waypoint execution - Implemented computer vision to detect QR and estimate relative pose using pixel-scaling methods (± 0.5 in accuracy) - Designed indoor navigation framework where UAV executed sequential command logic from encoded markers - Explored SLAM localization (ORB-SLAM3) and evaluated feasibility constraints due to software incompatibilities - Built foundational experience in UAV controls, perception, and autonomy through iterative testing and prototyping		
Avionics and Propulsion Engineer	SDSU Rocket Project	08/2023 – 07/2024
- Designed and fabricated instrumentation mounts (SOLIDWORKS + 3D printing) for pressure, temperature, and flow sensors integrated with NI DAQ systems - Built avionics ground system including sensor wiring, soldering, and data acquisition interfaces for propulsion testing - Developed thermally resilient enclosures for avionics hardware, improving survivability in high-temp test conditions - Investigated Venturi flow meter design for turbopump; developed Arduino-based pressure sensing architecture		
Instructional Student Assistant	SDSU College of Engineering	01/2025 – 12/2025
- Taught and reinforced orbital mechanics concepts (two-body motion, Keplerian orbits, transfers) for AE320 students - Taught and reinforced differential equations coursework for engineering applications for AE260 students		

SCHOOL PROJECTS**A E 403W – Senior Project | Presented at SDSU Senior Design 2026**

- Led development of a MATLAB/Simulink-based collision avoidance system for the NASA BAM Challenge
- Designed perception-prediction-planning pipeline to detect and avoid high-speed projectiles in real time
- Achieved 95% successful avoidance rate, with performance tradeoff leading to instability and crash behavior
- Evaluated system robustness under sensor uncertainty and noise perturbation

A E 460A + 460B – Spacecraft Design | Presented at SDSU Senior Design 2026

- Designed interplanetary trajectory using AGI STK, including Earth escape, Jupiter assist, and Solar Oberth maneuver
- Reduced mission ΔV from ~ 65 km/s baseline to 57.7 km/s through trajectory optimization
- Modeled multi-phase trajectory with gravitational perturbations and solar radiation pressure
- Developed mission timeline and validated trajectory segments for a 57-year deep-space mission architecture
- Built supporting MATLAB tools for initial mission architecture calculations

PROGRAMS

Mission Concept Academy (NASA L'SPACE)	05/2022 – 09/2022
- Performed thermal analysis and passive thermal control design for Mars rover systems - Designed dust-resistant heat dissipation system and thermally optimized payload housing - Collaborated in 12-person team developing end-to-end rover mission concept	
NASA Proposal Writing & Evaluation Experience (NASA L'SPACE)	01/2022 – 05/2022
- Developed engineering timeline and system-level project plan using Gantt methodology - Evaluated proposals using NASA scoring criteria and contributed to submission design	
NASA AFRC Engineering Design Challenge (NCAS Mission 3)	07/2022
- Led development of drone collision avoidance simulation using MATLAB/Simulink - Designed perception–prediction–control pipeline for detecting and avoiding high-speed projectiles - Evaluated performance under varying sensor noise conditions to assess robustness of guidance algorithm	

PERSONAL PROJECTS

Atmospheric Magnetoplasmodynamic (MPD) Ion Thruster built using a 1s Lipo battery, DC transformer, nickel strips, and an empty soda can.

YouTube channel “SciPK” where I explain science and engineering topics through video essays.

PUBLICATIONS

Khodadi, P., Cook, A., Tobita, Y., Liu, X., "AIAA SDSU Student Branch History," *AIAA SciTech 2026*, AIAA paper 2026-2727, Jan. 2026.

LEADERSHIP & ACTIVITIES

- Outreach Officer, AIAA SDSU – organized industry tours, panels, and events
- Membership Officer, SDSU Persian Student Association

CERTIFICATIONS

- **Ansys Systems Tool Kit (STK):** Level 3 Grand Master, Level 2 Master, Level 2 with Integrations, and Level 1.

AWARDS

- 2025-2026 Reuben H. Fleet Scholarship – AIAA San Diego
- 2026 Outstanding Contribution to the AIAA Student Chapter at SDSU

SKILLS

GNC Related

- AGI Systems Tool Kit (STK)
- Orbital mechanics
- Control System Design
- MATLAB / Simulink
- C/C++
- Python
- OpenCV
- Robot Operating System (ROS)
- Colosseum / AirSim
- Version Control (Git)
- Unreal Engine

Non-GNC Engineering

- Data Acquisition (DAQ)
- Soldering
- Wiring
- 3D Printing
- CAD (SOLIDWORKS and NX)
- Mission Concept Design
- JMARS
- Adobe Premiere Pro

Other Technical or Soft-Skills

- Public Speaking
- LaTeX (via Overleaf)
- Problem-Solving & Critical Thinking
- Teamwork & Collaboration
- Adaptability
- Attention to Detail
- Proposal Evaluation
- Bilingual: Persian, English